action-translation

AI-Powered Translation Automation for Lectures

GitHub Action + `translate` CLI • Claude Sonnet 5 • MyST Markdown

QuantEcon Project

What It Does

Automatically translates and reviews lectures when source content changes

Monitor merged PRs → Detect changes → Translate → Create PR → AI Review → Route

Key Capabilities

✓ **Smart Diff Translation** – Only translates modified sections

✓ **MyST Markdown Aware** – Preserves code, math, directives, anchors

✓ **Consistent Terminology** – Per-language glossaries (330–370 terms each)

✓ **Machine-Readable Review** – Verdict block with scores + auto-merge/editor routing

✓ **Deterministic Guardrails** – Structural parity, typography, bibliography backfill

✓ **Language Extensible** – zh-cn, fa, fr, ml, ja, es configured

How It Works

Section-Based Translation Approach

❌ Problem: Block-Level

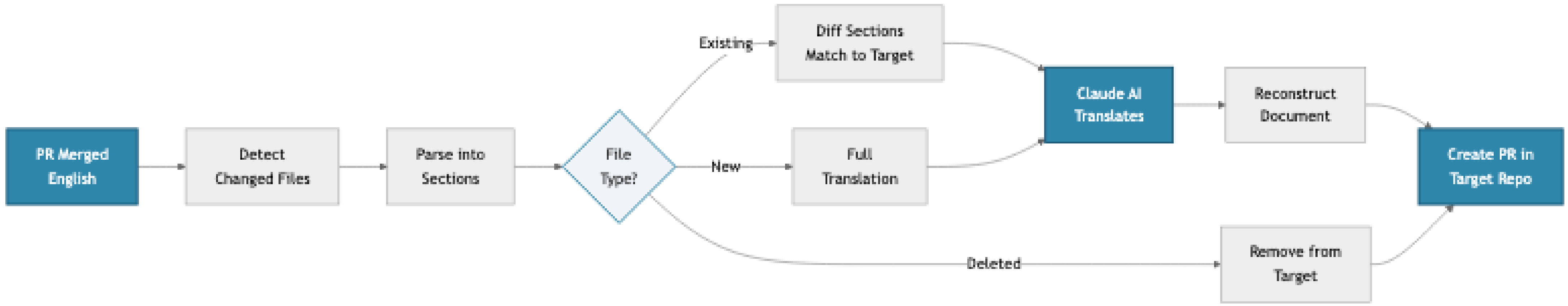
- Can't match across languages
- Loses translation context
- Complex mapping logic

✅ Solution: Section-Based

- Match by heading-map, then position
- Translate full sections
- Simple: Add, Update, Delete

Heading maps live in each translated file's frontmatter — English heading text mapped to its translation, so sections stay matched even after edits and reorders.

Translation Workflow



LLM-Powered Translation (Claude Sonnet 5)

UPDATE Mode

(changed sections)

- Sends: old EN + new EN + current translation
- Claude sees what changed
- Preserves style, terminology, and target-side localisation
- Glossary pins technical terms

NEW / RESYNC Modes

(new files, drift recovery)

- Whole-document translation with full context
- Deterministic post-pass: typography, heading map, frontmatter carry-forward
- Structural parity guard refuses corrupted output

Three Operational Modes

🔄 Sync Mode *(source repo)*

- Fires on merged PRs
- Translates changed sections
- Backfills cited `.bib` entries
- One PR per target language

🔀 Rebase Mode *(target repo)*

- Keeps sibling translation PRs current during waves

📝 Review Mode *(target repo)*

- Scores accuracy, fluency, terminology, formatting
- Deterministic diff checks (structure, heading map)
- Emits a machine-readable verdict block
- Routes: `auto-merge` vs `editor`
- Shadow mode records would-auto-merge without acting

The `translate` CLI

Operator tooling for everything the Action can't reach from CI

Command	Purpose
`init`	Bulk-seed a new edition from the source repo
`status`	Per-file sync state (`--check-sync` for content drift)
`forward`	Resync drifted files, one PR each (`--github`)
`backward`	Capture target-side improvements upstream
`review`	Local review of a translation PR
`setup` / `doctor`	Scaffold and verify workflow wiring

Trust & Safety

✓ **Trust-gated triggers** – resync commands require OWNER/MEMBER/COLLABORATOR

✓ **Fail-closed parsing** – malformed model output fails the run, never fabricates a verdict

✓ **Verdict provenance** – engine version + reviewed SHA in every verdict block

✓ **Append-only bibliography** – localised entries are never overwritten

✓ **Loud failure** – partial runs open issues; silent-success paths are being closed (W1)

Status & Getting Started

Current Status

📦 **v0.25.0** – deployed estate-wide
🔄 **Three modes**: Sync + Review + Rebase
✅ 1,500+ tests, type-checked, zero skips
🧪 27 E2E scenarios × 3 languages on real repos

In Production

- lecture-python.zh-cn
- lecture-python-programming (zh-cn / fa / fr)

Resources

GitHub: QuantEcon/action-translation

Docs: `docs/user/` (quickstart, tutorials, metadata contract)

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